```
  </channel>

  <image rdf:about="http://xml.com/universal/images/xml_tiny.
gif">
    <title>XML.com</title>
    <link>http://www.xml.com</link>
    <url>http://xml.com/universal/images/xml_tiny.gif</url>
  </image>

  <item rdf:about="http://xml.com/pub/2000/08/09/xslt/xslt.
html">
    <title>Processing Inclusions with XSLT</title>
    <link>http://xml.com/pub/2000/08/09/xslt/xslt.html</link>
    <description>
     Processing document inclusions with general XML tools
can be
     problematic. This article proposes a way of preserving
inclusion
      information through SAX-based processing.
    </description>
  </item>
</rdf:RDF>
```

Since the collection of RDF statements intrinsically represents a *labelled directed graph*, as a data model, it is more naturally suited to certain kinds of knowledge representation other than the relational model. When the data model imbues semantic Web applications, then RDF shines like in RSS, FOAF, conceptual graphs and topic maps. XUL uses RDF extensively for data binding.

It is remarkable that RDF database systems are the only standardised NoSQL solutions built on a simple, uniform data model and a powerful declarative query language for RDF graphs called SPARQL. Taking a resource-centric approach maps the RDF data model to object oriented systems and RESTful architectures.

NoSQL

Also referred to as 'Not only SQL', this is an umbrella term to describe a model of non-relational database systems. NoSQL data stores tend to be inherently distributed, schema-less and horizontally scalable.

A broad classification encompasses:

1. **Key value data stores:** The predominant data structure is a persistent hash table. Examples are BerkeleyDB, MemcacheDB, Tokyo Cabinet, Redis and SimpleDB.
2. **Document data stores:** These are key value stores, wherein the values are interpreted as semi-structured data and not a blob. Examples are CouchDB, MongoDB and Riak.
3. **Graph data stores:** These have a directed labelled graph model as the main data structure. Examples are neo4j, InfoGrid, HypergraphDB, AllegroGraph, 4store and Virtuoso.
4. **Wide column data stores:** Also known as an *extensible record set*, these have a column as the basic unit. A column is created for each row, and consists of a key and value pair. Usually the third entry is a time stamp. Examples are Apache Cassandra, HBase and SimpleDB.

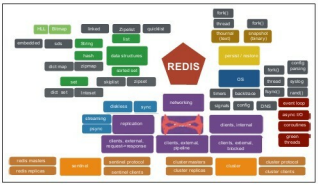

Figure 3: Redis data store and its modules

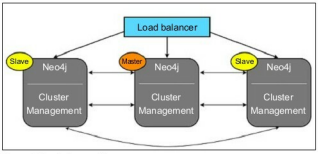

Figure 4: Neo4j clustered data store

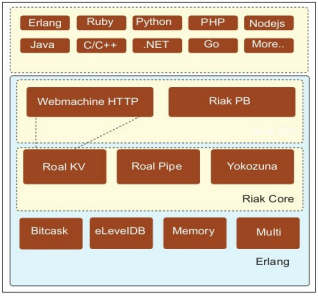

Figure 5: Riak architecture

Semi-structured data

Semi-structured data is a form of structured data where the data is annotated by tags, markers or headers to distinguish elements and enforce hierarchies of records and fields within the data.